Does Graph-Based Retrieval Beat Vector Search on Multi-Hop Questions?

A HippoRAG-style Study Using Personalized PageRank

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1. Dataset

2WikiMultiHopQA: Public dataset with annotated supporting passages, allowing retrieval recall to be measured against the provided evidence. We use a random sample of development questions and their corresponding passage corpus, with sample size and random seed fixed before the experiment.

2. Question

In RAG, vector search scores passages independently, so it can miss a second-hop passage connected through a shared entity. HippoRAG [1] instead builds an entity graph and uses Personalized PageRank (PPR). We test this claim at a small scale.

- **RQ1:** Does Graph + PPR achieve higher recall@2 and recall@5 than a dense-vector baseline?
- **RQ2:** With the same LLM reader, does the better retriever also achieve higher exact match (EM) and F1?

H1: Graph + PPR has higher retrieval recall, with the largest gains expected on questions requiring multiple passages.

H0: There is no difference in retrieval recall between the two methods. We report the result either way.

3. Tools

- **Baseline:** Dense embeddings indexed and searched using FAISS.
- **Graph + PPR:** An LLM extracts triples to build an entity graph, with synonym/entity matches generated using LSH over embeddings. Query entities are linked to graph nodes, PPR [4] runs from them, and the top-k passages are passed to the LLM reader.
- **Fair comparison:** Both methods use the same embedding model, passage corpus, and LLM reader.

Course topics: Graphs and Vectors, Hashing/LSH, Vector Search, Bootstrap and Hypothesis Tests.

4. Done Criteria

- **Evaluation:** Report recall@2, recall@5, all-recall, EM, and F1 for both methods. Include 95% bootstrap confidence intervals for recall, a paired Wilcoxon signed-rank test for recall, and McNemar's test for all-recall at 0.05.

Verdict rule: Graph + PPR is considered better if it shows a positive recall gain and the paired Wilcoxon test is significant at 0.05, with the 95% confidence interval reported alongside the estimate. Otherwise, we report no clear difference.

Main risks: Graph quality depends on LLM triple extraction and entity linking; the additional graph-building pipeline may affect retrieval independently of PPR; LLM cost is controlled with a fixed sample; and the single-dataset evaluation limits broad claims.

5. References

[1] B. J. Gutiérrez et al. "HippoRAG: Neurobiologically inspired long-term memory for large language models." NeurIPS, 2024.

[2] X. Ho et al. "Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps." COLING, 2020.

[3] P. Lewis et al. "Retrieval-augmented generation for knowledge-intensive NLP tasks." NeurIPS, 2020.

[4] T. H. Haveliwala. "Topic-sensitive PageRank." WWW, 2002.